

A Machine-Verified Disproof of TLMC Conjecture #578: Divisibility of Binomial-Matroid Characteristic Values

Batch 7 solution submission

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Abstract

We disprove a conjecture of the Last Math Competition asserting that for the binomial matroid $\text{bin}(n, i)$ — the binary matroid on the $\binom{n}{i}$ incidence vectors of weight i in $\mathbb{F}_2^n$, i.e. the i -subsets under symmetric difference (for $i = 2$ this is the graphic matroid $M(K_n)$) — the value $\chi_{n,i}(-m)$ of the characteristic polynomial is divisible by $i!(n-i)!$ for all $2 \leq i \leq n-2$ and $m \geq 1$. The smallest case already fails: $\chi_{4,2}(t) = t^3 - 6t^2 + 11t - 6$ and $\chi_{4,2}(-4) = -210$, while $2! \cdot 2! = 4 \nmid 210$. The computation is formalized in Lean 4 via Whitney's subset expansion with the linear rank computed by brute force over all 2^6 coefficient vectors, entirely kernel-verified by `decide`.

1 The conjecture

Conjecture 1.1 (TLMC-00000000578). Let $\text{bin}(n, i)$ be the binomial matroid (the i -subsets of $\{1, \dots, n\}$ under symmetric difference, i.e. the binary matroid of the weight- i vectors of $\mathbb{F}_2^n$) and $\chi_{n,i}$ its characteristic polynomial. For all $2 \leq i \leq n-2$ and integers $m \geq 1$, $\chi_{n,i}(-m)$ is divisible by $i!(n-i)!$, with quotient a nonnegative combination of Eulerian numbers.

2 The disproof

Definition 2.1 (Characteristic polynomial). For a matroid M on ground set E with rank function r , Whitney's subset expansion is

$$\chi_M(t) = \sum_{A \subseteq E} (-1)^{|A|} t^{r(E)-r(A)}.$$

For a binary matroid given by vectors, $r(A)$ is the $\mathbb{F}_2$ -dimension of the span of A , equivalently the maximum size of a linearly independent subfamily.

Theorem 2.2. *Conjecture 1.1 is **false**. At $n = 4$, $i = 2$ (the six weight-2 vectors of $\mathbb{F}_2^4$, i.e. the edges of K_4 , spanning the rank-3 even-weight subspace):*

$$\chi_{4,2}(t) = t^3 - 6t^2 + 11t - 6 \quad \left(= \frac{P_{K_4}(t)}{t} \right),$$

so $\chi_{4,2}(-4) = -64 - 96 - 44 - 6 = -210$, and $i!(n-i)! = 2! \cdot 2! = 4$ does not divide 210.

Proof. Kernel computation (Lean `decide`). The rank of each of the 64 subsets of the ground set is computed as the maximum size of a linearly independent subfamily, independence being checked by brute force over all 2^6 coefficient vectors of $\mathbb{F}_2^6$; the sanity theorem `linIndepB_sanity` exhibits the tester behaving correctly on $\emptyset$, singletons, an independent pair, and the dependent triple $\{v_0, v_1, v_3\}$ (where $v_0 + v_1 = v_3$). The subset expansion then yields the four coefficients of χ and its value at -4 . $\square$

Remark. The companion clause (the quotient being a nonnegative combination of Eulerian numbers) fails with the divisibility. For comparison, the cases $(n, i) = (5, 2)$ and $(7, 2)$ pass the divisibility for all $m \leq 39$ in our computation, so the failure at $(4, 2)$ is a genuine small-case refutation, not a misreading that fails everywhere.

3 Machine verification

`Batch7.lean` (namespace `TLMCBatch7`) contains the vector configuration `vec`, the computable independence tester `linIndepB` (with sanity checks), the rank function `rankOf`, the Whitney coefficients `chiCoeff`, and the refutation `t1m578_false` together with the packaged data `t1m578`. Compilation via `lake env lean Batch7.lean` exits 0 with zero errors and warnings (with `maxRecDepth` raised for the kernel computations); `#print axioms` reports at most `[propext, Classical.choice, Quot.sound]` (`t1m578_false` needs only `[propext]`); no `sorry` appears.